

Liam Tanner

liamtanner.dev | liam.tanner@brentwood.ca | linkedin.com/liam-tanner | github.com/liamtannerr

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, Linux Bash, JavaScript

Architecture & Logic: Data Structures, Algorithms, Object-Oriented Design, Distributed Systems, Concurrency

Cloud & Systems: AWS, Linux/UNIX Environments, Relational Databases, Docker, PostgreSQL

Tools & Automation: Git, CI/CD, Automated Testing Frameworks, Full SDLC Operations

Certifications: AWS Cloud Practitioner Essentials

EDUCATION

University of Victoria

Victoria, BC

Bachelor of Software Engineering

Sep 2021 – Aug 2026 (expected)

- 4.00/4.00 GPA - 93% Avg - 2025/2026 Session

- Relevant Coursework: Optimization for Machine Learning, Operating Systems, Databases, Data Mining

EXPERIENCE

Software Developer

Jun – Dec 2025

Natural Resources Canada

Victoria, BC

- Collaborated in an Agile Scrum team to design and implement a scalable ETL data transformation workflow, processing massive relational PostgreSQL datasets and reducing execution times from hours to seconds.
- Engineered an encrypted database release-versioning system using SQL to securely track metadata, ensuring full-lifecycle accountability and seamless feature rollouts across 2-week development sprints.
- Rebuilt legacy frontend infrastructure in JavaScript and wrote custom Linux Bash scripts to automate the deployment process for the entire engineering team.

Software Engineer

Jan 2025 – Aug 2026

AUVIC - Autonomous Underwater Vehicles Club

Victoria, BC

- Developed a ROS 2 dynamic configuration pipeline to extract PID and Kalman filter values from YAML files, eliminating C++ recompilation downtime and enabling real-time parameter tuning.
- Resolved critical thruster saturation limits and built a state machine for autonomous navigation, advancing the team to the semi-finals at the international RoboSub 2026 competition.

Research Assistant - Induced Seismicity Project

Sep 2023 – Apr 2024

Natural Resources Canada

Sidney, BC

- Contributed to the Induced Seismicity research project by developing complex CO2 injection simulations and waste-water disposal wells, utilizing Python meshes, and the OpenGeoSys finite element solver; becoming the first NRCan researcher to successfully model a real world site.

PROJECTS

AI Incident Orchestrator | *Python, PostgreSQL, CI/CD, LLM APIs*

May – Aug 2026

- Engineered an AI orchestration pipeline in a distributed environment, leveraging LLMs and RAG context to automate action recommendations for preventing detected cybersecurity threats.

Convolutional Engine | *C++, CUDA*

Apr 2026

- Built a high-performance image processing library from scratch in C++ and CUDA, implementing complex shared memory tiling algorithms to bypass global memory bottlenecks and maximize GPU thread throughput.

Breast Cancer Diagnosis Model | *MATLAB, Optimization Mathematics*

Feb 2026

- Engineered a regularized logistic regression model, applying optimization mathematics (regularized softmax cost function, gradient descent) from scratch to achieve 98.59% accuracy on a subset of the WDBC dataset.

Ray Tracer | *C++, Object-Oriented Design*

Mar – Jun 2025

- Developed a ray tracer capable of rendering 3D animations utilizing core vector math, abstract algorithms, and low-level rendering logic entirely from scratch.

Virtual Triage System | *Python, FastAPI, MongoDB, ReactJS*

Sep – Nov 2024

- Engineered a virtual triaging web application designed to allow prospective hospital, emergency room, patients to wait for an available physician from home.